

Big Data and Hadoop: An Overview

Data is being generated every moment of the day around the world, and can give organisations important insights into the acceptability of their products, among other things. Hadoop, a framework for distributed data analysis, is designed for the storage, processing and analysis of large datasets in a distributed computing environment. Let's see how it works with Big Data.



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Data is being generated at an unprecedented rate, which traditional data processing techniques can no longer handle. This is where Big Data comes into the picture — people, organisations, and machines generate huge amounts of structured and unstructured data. Big Data is crucial for businesses and organisations to gain insights into their operations, customer behaviour, and market trends. It can be used for applications such as

fraud detection, personalised marketing, predictive maintenance, and supply chain optimisation.

The role of a Big Data engineer is to design, build, and maintain the infrastructure required to store, process, and analyse large datasets. The increasing adoption of Big Data technologies is expected to create numerous job opportunities in the field. Engineers who specialise in Big Data can work in a variety of industries,

including finance, healthcare, retail, and technology. To excel in this field, one needs to be proficient in programming languages like Java, Python, and Scala. They must also possess knowledge of Hadoop, MapReduce, and be proficient in data modelling and warehousing.

Data is growing exponentially, and by 2025, it is estimated that 175 petabytes of data will be generated. In addition to social media, digital devices, online transactions, airlines,